SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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Department	CSE - AIDS	**Course Code**	DSA0604
Course Name	Data Handling and Visualization		

Employee Data Visualization Using R Programming

Activity 4 — R Programming Graphs, Code and Output

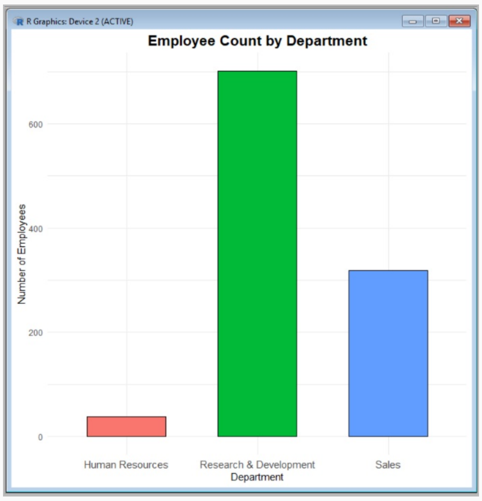
Graph 1: Employee Count by Department

CODE

```
library(ggplot2)

ggplot(data, aes(x = Department, fill = Department)) +
  geom_bar(color = "black", width = 0.6) +
  labs(
    title = "Employee Count by Department",
    x = "Department",
    y = "Number of Employees"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
    axis.text.x = element_text(size = 11),
    legend.position = "none"
  )
```

OUTPUT



Graph 2: Employee Count by Job Role

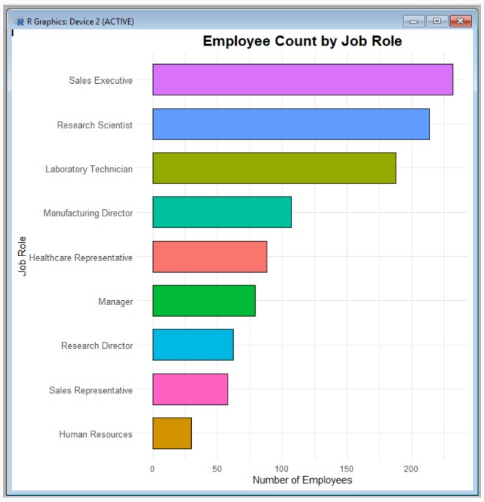
CODE

```
library(ggplot2)
library(dplyr)

job_count <- data %>%
  count(JobRole) %>%
  arrange(n)

ggplot(job_count, aes(x = reorder(JobRole, n), y = n, fill = JobRole)) +
  geom_col(color = "black", width = 0.7) +
  coord_flip() +
  labs(
    title = "Employee Count by Job Role",
    x = "Job Role",
    y = "Number of Employees"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
    axis.text.y = element_text(size = 10),
    legend.position = "none"
  )
```

OUTPUT



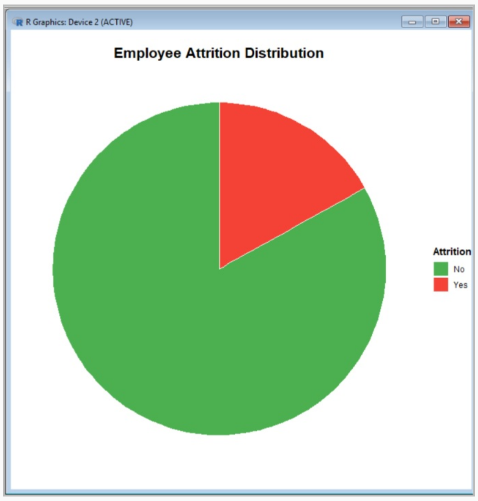
Graph 3: Employee Attrition Distribution

CODE

```
library(ggplot2)

ggplot(data, aes(x = "", fill = factor(Attrition))) +
  geom_bar(width = 1, color = "white") +
  coord_polar(theta = "y") +
  labs(
    title = "Employee Attrition Distribution",
    fill = "Attrition"
  ) +
  scale_fill_manual(
    values = c("#4CAF50", "#F44336"),
    labels = c("No", "Yes")
  ) +
  theme_void() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
    legend.title = element_text(face = "bold")
  )
```

OUTPUT



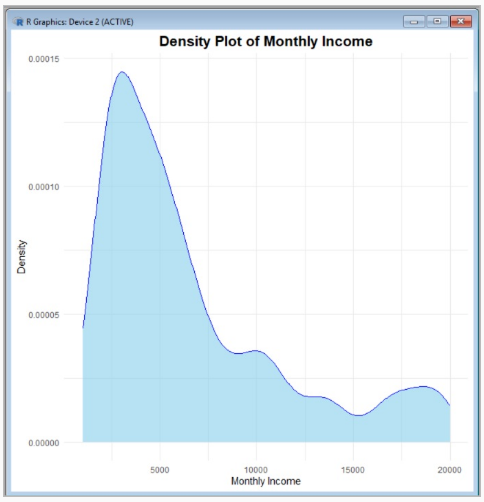
Graph 4: Monthly Income Distribution

CODE

```
library(ggplot2)

ggplot(data, aes(x = MonthlyIncome)) +
  geom_density(fill = "skyblue", color = "blue", alpha = 0.6) +
  labs(
    title = "Density Plot of Monthly Income",
    x = "Monthly Income",
    y = "Density"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16)
  )
```

OUTPUT



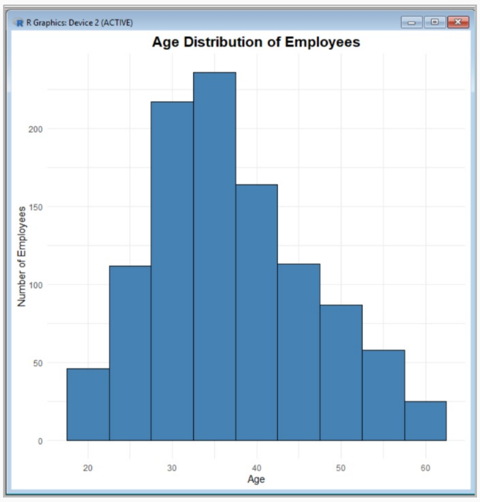
Graph 5: Age Distribution

CODE

```
library(ggplot2)

ggplot(data, aes(x = Age)) +
  geom_histogram(
    binwidth = 5,
    fill = "steelblue",
    color = "black"
  ) +
  labs(
    title = "Age Distribution of Employees",
    x = "Age",
    y = "Number of Employees"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16)
  )
```

OUTPUT



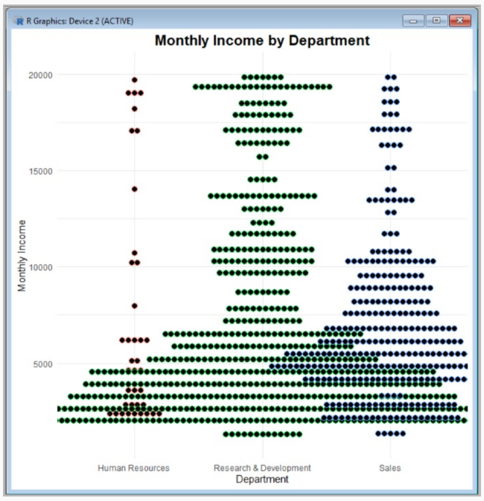
Graph 6: Monthly Income by Department (Dot Plot)

CODE

```
library(ggplot2)

ggplot(data, aes(x = Department, y = MonthlyIncome, color = Department)) +
  geom_dotplot(
    binaxis = "y",
    stackdir = "center",
    dotsize = 0.5
  ) +
  labs(
    title = "Monthly Income by Department",
    x = "Department",
    y = "Monthly Income"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
    legend.position = "none"
  )
```

OUTPUT



Graph 7: Average Monthly Income by Job Level

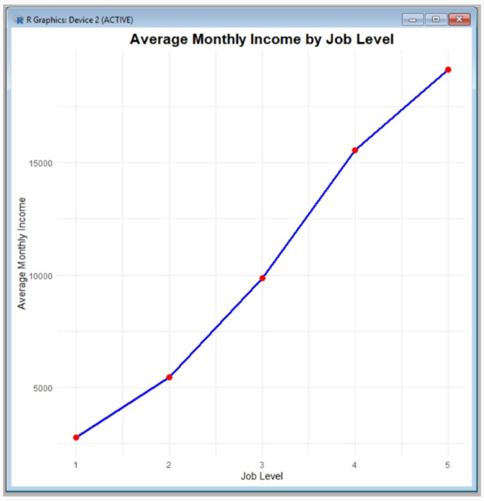
CODE

```
library(ggplot2)
library(dplyr)

line_data <- data %>%
  group_by(JobLevel) %>%
  summarise(AverageIncome = mean(MonthlyIncome))

ggplot(line_data, aes(x = JobLevel, y = AverageIncome)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Average Monthly Income by Job Level",
    x = "Job Level",
    y = "Average Monthly Income"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16)
  )
```

OUTPUT



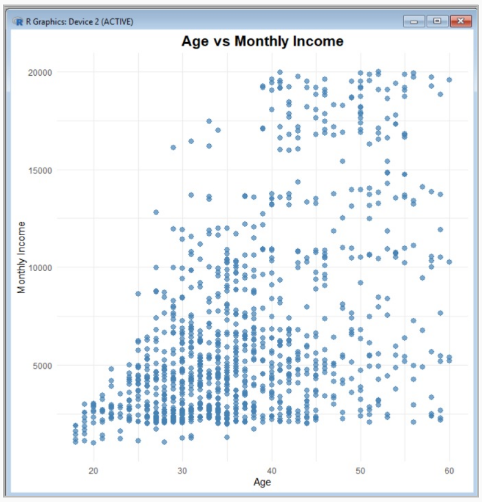
Graph 8: Age vs Monthly Income

CODE

```
library(ggplot2)

ggplot(data, aes(x = Age, y = MonthlyIncome)) +
  geom_point(
    color = "steelblue",
    size = 2.5,
    alpha = 0.7
  ) +
  labs(
    title = "Age vs Monthly Income",
    x = "Age",
    y = "Monthly Income"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 16
    )
  )
```

OUTPUT



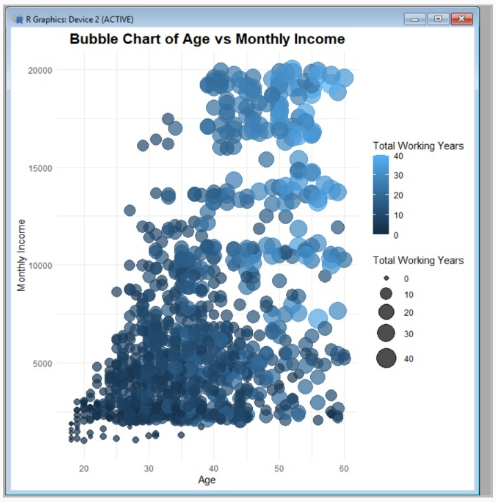
Graph 9: Age vs Monthly Income (Bubble Chart)

CODE

```
library(ggplot2)

ggplot(data, aes(
  x = Age,
  y = MonthlyIncome,
  size = TotalWorkingYears,
  color = TotalWorkingYears
)) +
  geom_point(alpha = 0.7) +
  scale_size(range = c(2, 10)) +
  labs(
    title = "Bubble Chart of Age vs Monthly Income",
    x = "Age",
    y = "Monthly Income",
    size = "Total Working Years",
    color = "Total Working Years"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold",
      size = 16
    )
  )
```

OUTPUT



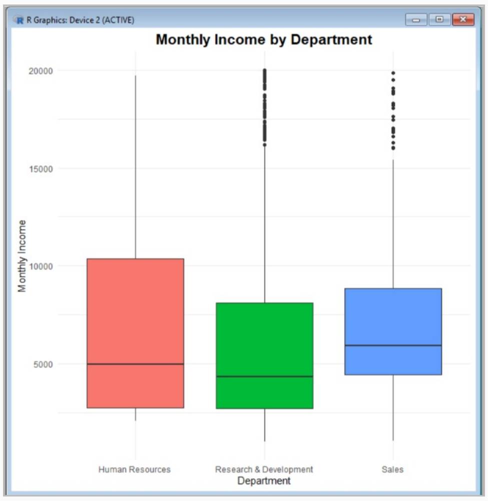
Graph 10: Monthly Income by Department (Boxplot)

CODE

```
library(ggplot2)

ggplot(data, aes(x = Department, y = MonthlyIncome, fill = Department)) +
  geom_boxplot() +
  labs(
    title = "Monthly Income by Department",
    x = "Department",
    y = "Monthly Income"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
    legend.position = "none"
  )
```

OUTPUT



Graph 11: Monthly Income by Job Role (Violin Plot)

CODE

```
library(ggplot2)

ggplot(data, aes(x = JobRole, y = MonthlyIncome, fill = JobRole)) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  labs(
    title = "Monthly Income Distribution by Job Role",
    x = "Job Role",
    y = "Monthly Income"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )
```

OUTPUT

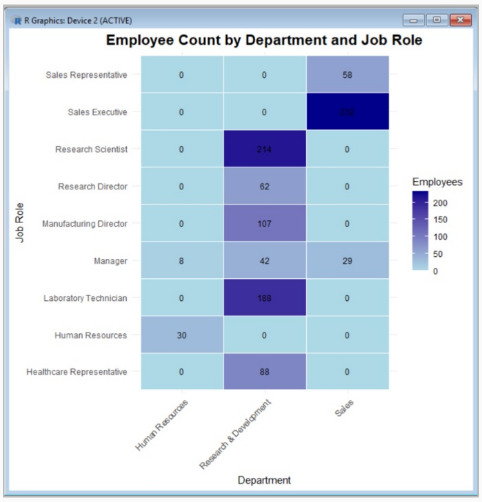
Graph 12: Department vs Job Role

CODE

```
library(ggplot2)

ggplot(data, aes(x = Department, y = JobRole)) +
  geom_bin2d() +
  scale_fill_gradient(low = "lightyellow", high = "red") +
  labs(
    title = "Employee Count by Department and Job Role",
    x = "Department",
    y = "Job Role",
    fill = "Count"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
```

OUTPUT



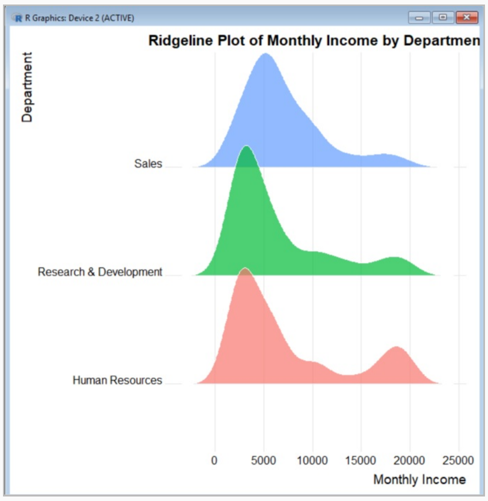
Graph 13: Monthly Income by Department (Ridgeline Plot)

CODE

```
library(ggplot2)
library(ggridges)

ggplot(data, aes(x = MonthlyIncome, y = Department, fill = Department)) +
  geom_density_ridges(alpha = 0.7, color = "white", scale = 1.2) +
  labs(
    title = "Ridgeline Plot of Monthly Income by Department",
    x = "Monthly Income",
    y = "Department"
  ) +
  theme_ridges() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
    legend.position = "none"
  )
```

OUTPUT



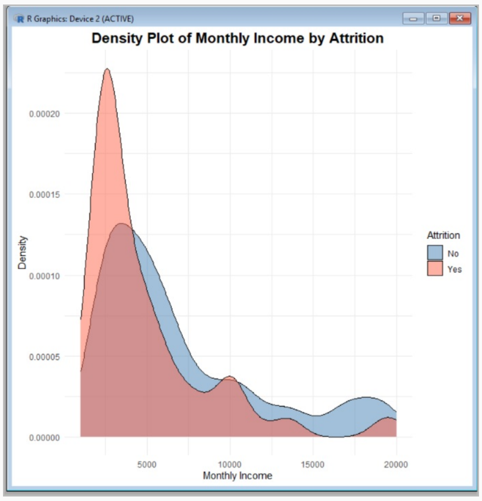
Graph 14: Monthly Income by Attrition Status

CODE

```
library(ggplot2)

ggplot(data, aes(x = MonthlyIncome, fill = factor(Attrition))) +
  geom_density(alpha = 0.5) +
  labs(
    title = "Density Plot of Monthly Income by Attrition",
    x = "Monthly Income",
    y = "Density",
    fill = "Attrition"
  ) +
  scale_fill_manual(
    values = c("steelblue", "tomato"),
    labels = c("No", "Yes")
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16)
  )
```

OUTPUT



Graph 15: Department → Job Role → Attrition (Sunburst)

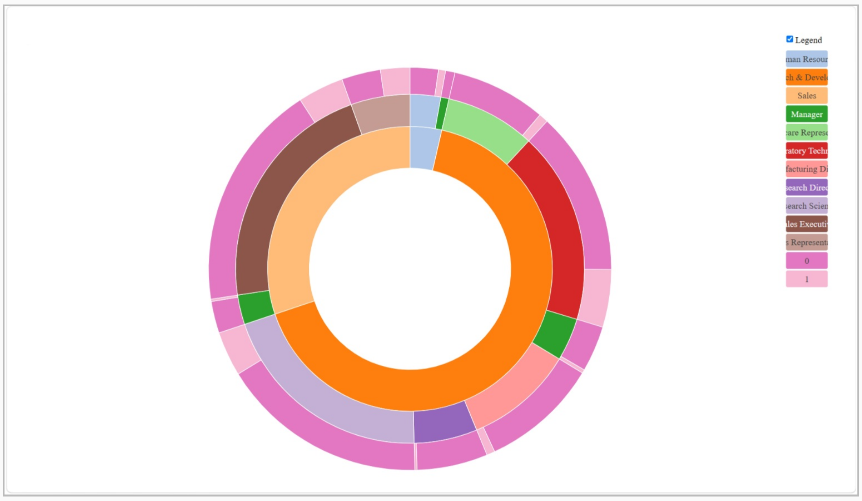
CODE

```
library(dplyr)
library(sunburstR)

sunburst_data <- data %>%
  count(Department, JobRole, Attrition) %>%
  transmute(
    sequence = paste(Department, JobRole, Attrition, sep = "-"),
    size = n
  )

sunburst(sunburst_data)
```

OUTPUT



— End of Activity 4 —